STEVENS INSTITUTE of TECHNOLOGY
School of Business

FE 610

Problem Set 2

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Spring 2025

For academic use only.

For the function $f(x) = \cos(x)$ find the first order variation from 0 to 2π.

For the function $g(x) = xe^x$ find the first order variation from -2 to 2.

We have a random walk with probability of getting a heads on any toss being $\frac{1}{3}$, where the random walk is defined as

$$L(n) = \sum_{i=1}^{n} Y_i, \text{ where } Y_i = \begin{cases} 3, & \omega_i = H \\ -1, & \omega_i = T \end{cases}$$

with $L(0) = 0$.

1. Let α be the first order variation of this walk up to time 4. Express α.
2. What is $\mathbb{E}[\alpha]$?
3. What is $\mathbb{V}(\alpha)$?

We have a random walk with probability of getting a heads on any toss being $\frac{1}{3}$, where the random walk is defined as

$$K(n) = \sum_{i=1}^{n} Z_i, \text{ where } Z_i = \begin{cases} 2, & \omega_i = H \\ -1, & \omega_i = T \end{cases}$$

with $K(0) = 0$.

1. Define β to be the first order variation of this process up to time 3. Express β.
2. What is $\mathbb{E}[\beta]$?
3. What is $\mathbb{V}(\beta)$?
4. Express the random variable $\gamma = [K, K](3)$
5. What is $\mathbb{E}[\gamma]$?
6. What is $\mathbb{V}(\gamma)$?

For this problem, we are going to use

$$X_i(\omega_i) = \begin{cases} 1 & \omega_i = H \\ -1 & \omega_i = T \end{cases}$$

Using this walk, we will define and work with two different walks which use the parameter n.

$$A^{(n)}(t) = \sum_{i=1}^{nt} \frac{1}{\sqrt[3]{n}} X_i$$

$$B^{(n)}(t) = \sum_{i=1}^{nt} \frac{1}{n} X_i$$

Answer the following:

1. Express the first order variation and quadratic variation of both walks.
2. What is the limit of the first order and quadratic variation of each walk as $n \to \infty$?

For a Brownian motion $W(t)$, answer the following:

1. Express $\mathbb{P}(W(t) > 4)$
2. Find $\mathbb{P}(W(5) > 2)$
3. If we define the process $X(t) = e^{W(t)}$, is this process a martingale?

For a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, define $\Omega = \{a, b, c, d\}$ and let $\mathbb{P}(a) = \frac{1}{12}$, $\mathbb{P}(b) = \frac{1}{8}$, $\mathbb{P}(c) = \frac{1}{6}$, and $\mathbb{P}(d) = \frac{15}{24}$. Define the random variables:

$$X(\omega) = \begin{cases} 1, & \omega \in \{a, b\} \\ 0, & \omega \in \{c, d\} \end{cases}$$

$$Y(\omega) = \begin{cases} -2, & \omega \in \{a, c\} \\ -1, & \omega \in \{b, d\} \end{cases}$$

1. What is $\mathbb{E}[X|Y]$ and $\mathbb{E}[Y|X]$?
2. Define $Z = 2X + 4Y$. What is $\mathbb{E}[Z|Y]$ and $\mathbb{E}[Z|X]$?